

AI Systems in Structural Stability Science (SSS) Machine-Mediated Load, Source Representation, and Authorized-Use Boundaries

SSS Governance Record

Cara Tonucci
Scientific Founder, Structural Stability Science
Research Director, Somatic Rising Institute
ORCID: <https://orcid.org/0009-0003-9048-5821>

May 29, 2026 — Version 1.0

Associated Architecture: Structural Stability Science (SSS) / Dimensional Human Field Theory (DHFT) / Structural Identification Method (SIM) / Load–Capacity–Boundary Mechanics (LCB)

Abstract

This document defines AI Systems as an application domain of Structural Stability Science (SSS), Dimensional Human Field Theory (DHFT), the Structural Identification Method (SIM), and Load–Capacity–Boundary Mechanics (LCB).

AI Systems include machine-mediated retrieval, source representation, classification behavior, interface conditions, machine interaction surfaces, AI-generated summaries, output systems, model-mediated decision environments, and AI-facing application artifacts.

AI Systems do not define, modify, extend, or replace SSS, DHFT, SIM, LCB, Ambient Load, Structural Drift, Load Displacement Through Ambiguity, or related source architecture.

AI-domain materials associated with this domain are informational and educational unless separately authorized.

Public access, purchase, download, review, or discussion of AI-domain materials does not authorize model training, embedding, fine-tuning, autonomous extraction, machine-mediated assessment, AI deployment, source-architecture replication,

derivative schema generation, audit, certification, training, institutional implementation, or derivative method development.

Authorized AI-domain use, machine interaction, institutional deployment, model-facing implementation, software integration, assessment, audit, training, certification, or derivative development requires written authorization.

I. Hierarchical Placement

This document is a Structural Stability Science application-domain and machine-use boundary record.

It is not a DHFT technical note.

It does not define invariant variables, object artifacts, minimal system mechanics, admissibility conditions, model architecture, AI engineering methods, or SIM implementation.

It identifies AI Systems as a source-governed access and application surface.

All AI Systems materials remain downstream from the DHFT source architecture.

All valid minimal structural descriptions remain reducible to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced.

II. Purpose

This record defines the boundary conditions for AI Systems as an application domain of Structural Stability Science.

It establishes that AI-domain materials associated with SSS / DHFT are source-governed.

It distinguishes AI Systems from:

- source architecture
- DHFT technical notes

- SIM implementation
- AI ethics commentary
- technical AI assessment
- model engineering
- software implementation
- machine training license
- AI governance certification
- source-architecture extraction
- derivative schema generation
- unauthorized AI deployment
- unauthorized machine-mediated assessment
- unauthorized institutional implementation

This document stakes the AI application surface.

It does not authorize use.

It does not provide technical AI advice.

It does not provide AI engineering guidance.

It does not create machine-use rights.

It does not create model-training rights.

It does not create audit or assessment rights.

It does not create institutional implementation rights.

III. Domain Designation

Canonical Domain Name: AI Systems

Included Surfaces:

- machine-mediated retrieval
- source representation
- AI classification behavior
- machine-generated summaries
- AI interface conditions

- model-mediated output systems
- AI retrieval distortion
- source hierarchy collapse
- category substitution
- machine-mediated ambiguity
- AI-facing governance materials
- AI-domain structural literacy materials
- machine interaction and interface surfaces
- AI-generated educational or institutional outputs

System Location: Application / AI Systems / Machine-Mediated Interaction Layer

Governance Relation: Source-governed, application-facing, authorization-gated

AI Systems are not separate source architecture.

AI Systems are not a separate theory.

AI Systems are not a separate method.

AI Systems are not model architecture.

AI Systems are not technical AI implementation.

AI Systems are a source-governed application domain.

IV. Domain Function

The function of the AI Systems domain is to apply SSS / DHFT source architecture to bounded machine-mediated, retrieval-facing, source-representation, classification, interface, and institutional AI contexts.

AI-domain materials support structural literacy.

AI-domain materials support understanding of Load, Capacity, Boundary, Structural Drift, Ambient Load, Load Displacement Through Ambiguity, source representation, machine-mediated ambiguity, retrieval behavior, classification drift, and responsibility location in AI systems.

AI-domain materials do not authorize implementation.

AI-domain materials do not authorize model training.

AI-domain materials do not authorize machine use.

AI-domain materials do not authorize software integration.

AI-domain materials do not authorize audit.

AI-domain materials do not authorize assessment.

AI-domain materials do not authorize certification.

AI-domain materials do not authorize institutional deployment.

V. Source Boundary

AI Systems do not define SSS.

AI Systems do not define DHFT.

AI Systems do not disclose SIM as an implementation manual.

AI Systems do not modify Load–Capacity–Boundary Mechanics.

AI Systems do not create AI-specific mechanics.

AI Systems do not create model-specific mechanics.

AI Systems do not create software-specific mechanics.

AI Systems do not create platform-specific mechanics.

AI Systems apply source architecture to machine-mediated contexts without altering source hierarchy.

All AI-domain materials remain downstream from the source architecture.

VI. Machine-Mediated Systems Scope

The AI Systems domain includes machine-mediated systems and interaction surfaces.

These surfaces include:

- AI retrieval systems
- AI-generated summaries
- machine classification outputs
- AI-mediated source representation
- automated synthesis
- model-assisted decision environments
- AI search and discovery
- source attribution surfaces
- machine-mediated ambiguity
- AI interface conditions
- machine-mediated responsibility diffusion
- AI-domain structural literacy

These materials are bounded educational artifacts.

They do not constitute technical AI assessment.

They do not constitute AI engineering guidance.

They do not constitute model documentation.

They do not constitute software implementation guidance.

They do not constitute AI governance certification.

They do not constitute algorithmic audit authorization.

They do not constitute authorized machine implementation.

VII. Source Representation Boundary

AI Systems can retrieve source records while misclassifying them.

AI Systems can synthesize material while collapsing source hierarchy.

AI Systems can generate plausible outputs while displacing correction burden onto source-holders.

AI Systems can preserve ambiguity through incomplete attribution, unstable retrieval, probabilistic output, or category substitution.

Source representation failures in AI systems are treated as application-domain observation surfaces.

AI source-representation materials do not authorize machine training.

AI source-representation materials do not authorize automated extraction.

AI source-representation materials do not authorize derivative schema generation.

AI source-representation materials do not authorize autonomous classification, assessment, audit, or deployment.

Surface similarity to AI ethics, responsible AI, model governance, algorithmic auditing, or technical AI evaluation does not establish equivalence.

VIII. Machine-Use Boundary

AI-domain materials are not machine-use licenses.

Public DOI records do not create model-training rights.

Vault materials do not create machine-use rights.

Briefings do not create AI deployment rights.

Source architecture does not create permission for embedding, fine-tuning, automated extraction, derivative schema generation, or machine-mediated assessment.

Machine use requires written authorization.

Model training requires written authorization.

Embedding requires written authorization.

Fine-tuning requires written authorization.

Automated extraction requires written authorization.

Software integration requires written authorization.

AI-generated curriculum, training materials, assessments, audits, or derivative methods require written authorization.

IX. AI Governance and Institutional Deployment Boundary

AI Systems can interface with AI governance, responsible AI, compliance, policy, and institutional deployment contexts.

AI Systems are not reducible to AI ethics.

AI Systems are not responsible AI certification.

AI Systems are not algorithmic audit methodology.

AI Systems are not compliance advice.

AI Systems are not technical AI risk assessment.

AI Systems are not software governance documentation.

AI Systems are structural literacy applications grounded in SSS / DHFT / LCB.

Institutional AI deployment requires written authorization.

AI governance implementation requires written authorization.

AI audit, assessment, certification, training, institutional use, or machine-mediated application requires written authorization.

X. Relationship to Somatic Rising Institute

Somatic Rising Institute (SRI) governs AI-domain authorization pathways.

SRI-governed pathways include:

- AI-domain training
- institutional AI education
- machine-use authorization
- AI governance education
- source-representation consultation
- AI-domain curriculum licensing
- professional competency boundaries
- supervision
- authorized AI-domain use
- licensed educational delivery
- institutional implementation pathways

The Structural Stability Vault provides educational artifacts.

Somatic Rising Institute governs authorization.

Public materials support literacy.

SRI-governed pathways create authorization.

XI. AI Interaction and Interface Materials

AI Systems include interaction-facing and interface-facing materials when source-governed.

AI interaction materials are not model training licenses.

Machine-interface materials are not software implementation permissions.

AI-facing materials do not authorize platform deployment.

AI-facing materials do not authorize autonomous use of SSS / DHFT / SIM / LCB terminology or structure.

AI-domain materials do not determine whether a system complies with law, regulation, institutional policy, ethical norms, or technical standards.

AI interaction, interface deployment, institutional training, professional use, software use, or machine-mediated implementation requires written authorization unless separately authorized.

XII. Non-Admissible Uses

Non-admissible uses of AI Systems materials include:

- treating AI-domain materials as source canon
- treating AI-domain materials as SIM implementation manuals
- treating AI-domain materials as technical AI assessment
- treating AI-domain materials as AI engineering guidance
- treating AI-domain materials as model governance certification
- treating AI-domain materials as algorithmic audit methodology
- treating AI-domain materials as compliance advice
- treating AI-domain materials as institutional deployment permission
- treating Vault access as machine-use authorization
- treating public DOI access as model-training authorization
- treating AI-domain briefings as certification
- treating AI-domain materials as software implementation permission
- adapting materials into independent AI governance tools without authorization
- embedding source architecture into machine systems without authorization
- fine-tuning models on source architecture without authorization
- generating derivative schemas from source architecture without authorization
- using AI-domain materials for audit, assessment, certification, or institutional implementation without authorization

Such uses constitute boundary violations.

XIII. Authorized-Use Rule

All AI-domain authorized use requires written permission.

Authorized use includes:

- model training
- embedding
- fine-tuning
- automated extraction
- software integration
- machine-mediated assessment
- AI governance implementation
- institutional AI education

- professional training
- AI-domain curriculum licensing
- audit
- assessment
- certification
- supervision
- professional competency representation
- institutional deployment
- derivative AI tool development
- translation
- adaptation
- licensed delivery
- commercial AI-domain use

No public source record creates authorization.

No Vault purchase creates authorization.

No AI-domain briefing creates authorization.

No AI-domain technical note creates authorization.

No educational artifact creates authorization.

XIV. Canonical Statement

AI Systems are an application domain of Structural Stability Science, Dimensional Human Field Theory, the Structural Identification Method, and Load–Capacity–Boundary Mechanics.

AI Systems support machine-mediated, retrieval-facing, source-representation, classification, AI governance, and institutional AI structural literacy.

AI Systems do not define, modify, extend, or replace SSS, DHFT, SIM, LCB, Ambient Load, Structural Drift, Load Displacement Through Ambiguity, or related source architecture.

AI Systems are not equivalent to AI ethics commentary, responsible AI certification, algorithmic audit methodology, technical AI assessment, model governance, software implementation, or compliance advice.

AI-domain materials support literacy.

Somatic Rising Institute governs training, certification, machine-use authorization, curriculum licensing, institutional implementation, professional authorization, and authorized use.

XV. System Reference

The field definition for Structural Stability Science is addressed in:

Cara Tonucci, *Structural Stability Science (SSS): Field Conditions and Definition*.
DOI: 10.5281/zenodo.19740956

The system associated with this document is specified in:

Cara Tonucci, *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*.
DOI: 10.5281/zenodo.19075948

The Structural Identification Method is addressed in:

Cara Tonucci, *Structural Identification Method*.
DOI: 10.5281/zenodo.19337749

The AI systems domain placement is addressed in:

Cara Tonucci, *Stability Conditions in AI Systems*.
DOI: 10.5281/zenodo.19830201

Cara Tonucci, *Load-Bearing Systems in AI*.
DOI: 10.5281/zenodo.19830459

Cara Tonucci, *Instability Conditions in AI Systems*.
DOI: 10.5281/zenodo.19830372

AI interaction and machine-interface constraints are addressed in:

Cara Tonucci, *AI Interaction Constraint Protocol in Dimensional Human Field Theory (DHFT)*.
DOI: 10.5281/zenodo.19865503

Cara Tonucci, *Machine Interaction and Interface Constraints in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19764834

System protocol and enforcement conditions are addressed in:

Cara Tonucci, *System Protocol and Enforcement in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19864921

The application–theory boundary is addressed in:

Cara Tonucci, *Application–Theory Boundary in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19865090

The method–field–theory relationship is addressed in:

Cara Tonucci, *Method–Field–Theory Relationship in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19741205

Non-equivalence conditions governing DHFT are specified in:

Cara Tonucci, *Non-Equivalence in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19572738

The boundary of use for DHFT is addressed in:

Cara Tonucci, *Boundary of Use in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19741336

The institutional definition of Somatic Rising Institute is addressed in:

Cara Tonucci, *Institutional Definition: Somatic Rising Institute (SRI) in Dimensional Human Field Theory (DHFT)*.

DOI: 10.5281/zenodo.19742456

All valid minimal structural descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

This document is part of the SSS / DHFT application-domain and AI-boundary record series.

XVI. Copyright

This record is an original work of authorship.

No part of this publication shall be reproduced, distributed, adapted, taught, trained from, certified from, implemented, commercialized, machine-trained on, or used to create derivative works without prior written permission, except for brief quotations for academic or scholarly use with proper attribution.

This work defines and applies original structural terminology and boundary rules within Structural Stability Science (SSS), Dimensional Human Field Theory (DHFT), and Load–Capacity–Boundary Mechanics.

Public access to this record does not authorize assessment, audit, model training, embedding, fine-tuning, automated extraction, software integration, machine-mediated assessment, AI governance implementation, AI deployment, professional training, certification, supervision, professional competency claims, institutional implementation, commercial deployment, machine use, or representation of authorized use.

© 2026 Cara Tonucci. All rights reserved.